

Notes of the course Introduction to Quantum Computing
by Gianna M. Del Corso

based on first draft by: Federico Matteoni

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Capitolo 1

Lecture 3

In this introductory lecture, we cover essential concepts in the realm of quantum computing. In particular

- Complex Numbers (Cartesian and polar forms, Euler's formula, and Euler's identity. Roots of unity, the primitive n th root of unity and its properties).
- Hilbert spaces and quantum state spaces, quantum orthonormal bases and visualization on the Bloch sphere.
- Single qubit gates (Pauli matrices, Hadamard Gate, other singular qubit gate).

1.1 Complex Numbers

In this introductory chapter, we will review basic concepts on complex numbers, touching on key concepts such as the Cartesian and polar forms, roots of unity, and Euler's formula. These concepts serve as the foundation for understanding quantum computing operations and algorithms.

Complex numbers are an extension of the real number system, introducing the imaginary unit denoted as i , where $i^2 = -1$. They can be represented in a 2D Cartesian coordinate system, with the real part, often denoted as $\Re(z)$, on the x -axis and the imaginary part, $\Im(z)$, on the y -axis.

1.1.1 Cartesian and Polar Forms

Any complex number $z \in \mathbb{C}$ can be expressed in both the Cartesian and polar forms. The Cartesian form is given as:

$$z = a + ib,$$

where a and b are real numbers.

In the polar form, z is represented as:

$$z = \rho(\cos \theta + i \sin \theta) = \rho \cdot e^{i\theta},$$

where ρ represents the distance from the origin, and θ is the angle that z forms with the positive real-axis. ρ is known as the modulus, and $e^{i\theta}$ is referred to as the phase.

The modulus of z is defined as:

$$|z|^2 = z \cdot z^* = a^2 + b^2 = \rho^2 = |z^*|^2,$$

where $z^* = \bar{z} = a - ib$ is the complex conjugate of z .

The Euler's Identity Euler's formula is a profound mathematical result that connects complex exponentials with trigonometric functions.

In analogy with the exponential function over the reals, we can define $e^z, z \in \mathbb{C}$, by means of the power series:

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad z \in \mathbb{C}.$$

When z is purely imaginary, $z = ix$ for some real x , we obtain that

$$e^{ix} = \cos x + i \sin x.$$

Euler's identity is a remarkable result obtained when evaluating previous formula for $x = \pi$:

$$e^{i\pi} = -1 \quad \text{or} \quad e^{i\pi} + 1 = 0.$$

This identity unites essential mathematical constants, including the imaginary unit, two irrational numbers, and the integers 1 and 0.

1.2 Roots of Unity

In the context of quantum computing, we often encounter equations of the form $z^n - 1 = 0$. These equations have n complex solutions in $\mathbb{C}$, which are referred to as the n th roots of unity. Among these roots, the primitive n th root of 1 is expressed as:

$$\omega_n = e^{\frac{2\pi i}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}.$$

For example, when $n = 3$, we have $\omega_3 = \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi = \frac{1}{2} + i \frac{\sqrt{3}}{2}$, which is a solution of $z^3 - 1 = 0$.

The roots of unity possess fascinating properties and can be generated from the primitive root taking all the powers from 1 to n

$$\omega_n^k = \left(\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \right)^k = \cos \frac{2\pi}{n} k + i \sin \frac{2\pi}{n} k = e^{\frac{2\pi i k}{n}}, \quad k = 1, \dots, n.$$

Other intriguing properties of the roots of unity include:

$$\omega_n^0 = 1$$

$$\omega_n^{n+k} = \omega_n^n \cdot \omega_n^k = \omega_n^k$$

$$\omega_n^{n-k} = \omega_n^n \cdot \omega_n^{-k} = \omega_n^{-k}$$

When working with complex numbers and quantum computing, it's often more convenient to represent them in polar form, as this simplifies multiplication, division, and exponentiation.

1.3 Hilbert Spaces

In the realm of quantum computing, we build our foundation upon Hilbert spaces. It's worth noting that for quantum computing, we primarily work with finite-dimensional Hilbert spaces, which are essentially complex vector spaces equipped with an inner product.

An inner product, denoted as $\langle \cdot | \cdot \rangle$, is a key concept in quantum mechanics and quantum computing. It is a map that takes two vectors from the Hilbert space $\mathbb{H}$ and returns a complex number, $\langle \cdot | \cdot \rangle : \mathbb{H} \times \mathbb{H} \rightarrow \mathbb{C}$. This inner product follows a set of important properties:

1. $\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$ (Hermitian symmetry).
2. $\langle \phi | \phi \rangle \geq 0$ with equality only when $\phi = 0$.
3. $\langle \psi | \alpha \phi_1 + \beta \phi_2 \rangle = \alpha \langle \psi | \phi_1 \rangle + \beta \langle \psi | \phi_2 \rangle$ (linearity).

Quantum state spaces are a fundamental concept in quantum mechanics and serve as the mathematical foundation for understanding and describing the quantum behavior of particles and systems. They play a crucial role in quantum computing, quantum information theory, and quantum algorithms, where they represent the possible quantum states of qubits and quantum systems.

1.3.1 Quantum State Spaces and Qubits

Definition 1 *A quantum state space is a complex vector space equipped with an inner product. In this space, each vector represents a possible state of a quantum system. Quantum states are typically represented as normalized vectors, meaning that their norm (length) is equal to 1.*

The fundamental unit of quantum information is the qubit. The qubit is the quantum counterpart of the classical bit, and it can be seen as a 2-dimensional quantum state space.

For instance consider a system with two basis states that we call $|0\rangle$ and $|1\rangle$. We can associate to the two orthonormal vectors of $\mathbb{C}^2$, for example $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. So a qubit is simply a linear combination of the basis states,

$$\alpha_0 |0\rangle + \alpha_1 |1\rangle,$$

where the amplitudes α_i are complex numbers such that $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

Similarly, in a 4-dimensional state space, we have a computational basis with vectors represented as:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad |2\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |3\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

and we need two qubits to represent a quantum state in a 4 dimensional state space.

A quantum state is always a vector of unit length in a quantum state space. In other words, the norm of a quantum state is 1, represented as $\| |v\rangle \| = 1$, and its norm is calculated as $\|v\| = \langle v|v\rangle$, where the inner product is usually the Euclidean inner product of $\mathbb{C}^n$.

We often use the bra-notation as well:

$$\langle v| = (v_0^*, \dots, v_{d-1}^*)$$

$$\langle v|w\rangle = \langle v|w\rangle \in \mathbb{C}.$$

It's important to distinguish the bra-ket notation from the ket-bra notation, where $|v\rangle \langle w|$ gives a rank-one matrix (a column-by-row matrix).

1.4 Quantum Basis and Bloch Sphere

A key feature of quantum systems is the flexibility in choosing different bases. Working with orthonormal bases (ONB) is particularly useful. For instance:

$$|0\rangle = |\uparrow\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = |\downarrow\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Another ONB can be obtained using $|+\rangle$ and $|-\rangle$:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

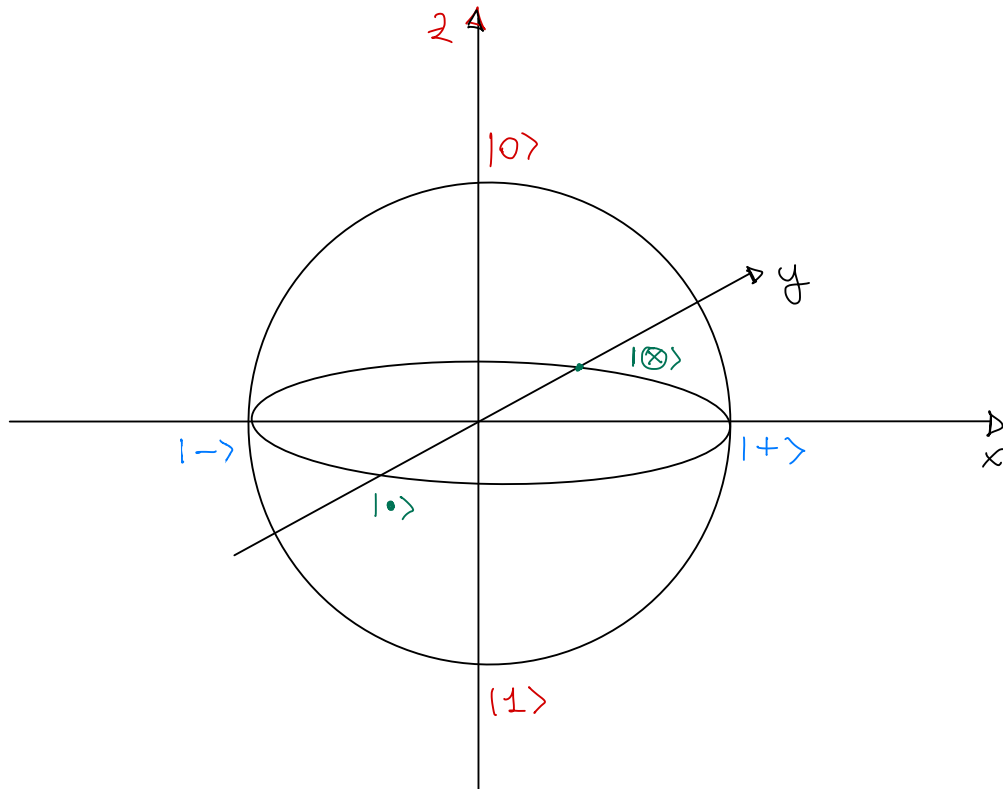
We can also define spin-in and spin-out states:

$$|\otimes\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$|\bullet\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

An interesting property of quantum states is that multiplying them by a global phase ($e^{i\theta}$) doesn't change the nature of the state. A global phase multiplies all components of the basis vector and doesn't affect the probabilities of measurement.

The **Bloch sphere** provides an intuitive way to visualize single-qubit states. Qubits are constrained to the surface of the Bloch sphere, with each point on the sphere representing a quantum state.



Bloch sphere with the main orthonormal basis

1.5 Pauli Matrices

Pauli matrices are 2x2 matrices in the world of quantum computing. They are essential for describing transitions between different bases. These matrices are denoted as σ_j (with $j \in \{x, y, z\}$) or as capital Roman letters X, Y, Z . The Pauli matrices perform 180-degree rotations around their corresponding axes.

The Pauli matrices are:

- $X = \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0|$. The X matrix is known as the bit-flip operator.

It essentially swaps the states $|0\rangle$ and $|1\rangle$.

- $Y = \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$. The Y matrix is often referred to as the bit-and-phase flip operator. It swaps $|0\rangle$ and $|1\rangle$ and introduces complex phase factors.

- $Z = \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$. The Z matrix is the phase-flip operator. It inverts the phase of $|1\rangle$ while leaving $|0\rangle$ unchanged.

The action of these matrices results in a reflection along the corresponding axis. For example:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle$$

This means that the points on the x -axis stay almost the same. Notably, $|+\rangle$ and $|-\rangle$ are not affected by this transformation.

- $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

σ_x leaves $|+\rangle$ unchanged and maps $|-\rangle$ to $-|-\rangle$, demonstrating that the vectors differ only by a global phase.

Similarly, σ_y performs a 180-degree rotation around the y -axis, while σ_z rotates around the z -axis.

The Pauli matrices possess an interesting property: they anti-commute. In other words, the following theorem holds:

Theorem 1.5.1 *For any pair of Pauli matrices σ_i and σ_j , where $i \neq j$ are elements from the set $\{x, y, z\}$, we have:*

$$\sigma_i \sigma_j = -\sigma_j \sigma_i.$$

Additionally,

$$\sigma_i^2 = I.$$

This property underscores the significance of Pauli matrices in quantum mechanics.

1.6 The Hadamard Matrix

The Hadamard matrix is a vital rotation matrix in quantum computing. It plays a significant role in creating superposition between states like $|+\rangle$ and $|-\rangle$.

The Hadamard matrix is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

This matrix allows us to transition from the canonical basis to the $\{|+\rangle, |-\rangle\}$ basis, where:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = |+\rangle = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = |\leftarrow\rangle = |-\rangle$$

Notably, H is such that $H^2 = I$, meaning that its inverse is itself ($H^{-1} = H$). This means that applying H twice gets us back to the original state.

For example, H maps $|0\rangle$ to $|+\rangle$, and applying H again to $|+\rangle$ takes us back to $|0\rangle$, and the same holds for $|1\rangle$ and $|-\rangle$. Indeed $H|+\rangle = \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle) + \frac{1}{2}(|0\rangle - |1\rangle) = |0\rangle$, we have that the positive and negative amplitudes of $|1\rangle$ cancel out. We have the so called interference a phenomenon often exploited in algorithms.

The Hadamard matrix is also a rotation about the y -axis by 90 degrees, followed by a rotation about the x -axis by 180 degrees, i.e. $H = e^{i\phi}XY^{1/2}$, with $\phi = -\pi/4$, and $Y^{1/2} = \frac{1}{2}(1+i) \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

1.7 Other Common Single Qubit Gates

Apart from the Pauli matrices (X, Y, Z) and the Hadamard matrix (H), there are other common single-qubit operations in quantum computing. These include:

- $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $S = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
- $T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

It's essential to note that there are infinitely many single-qubit quantum gates, but any arbitrary gate can be broken down into a product of rotations, which are matrices of the form:

$$\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

and matrices of the form:

$$\begin{bmatrix} e^{-i\beta/2} & 0 \\ 0 & e^{i\beta/2} \end{bmatrix}.$$

1.8 Exercises

Exercise 1.1

Write in polar form the following complex numbers written in Cartesian form

$$2 + i, -1 + 2i, \frac{\sqrt{3}}{2} - \frac{1}{2}i.$$

Write in Cartesian form the following complex numbers written in polar form

$$\sqrt{2}e^{\pi/2i}, -e^{-\pi/8i}.$$

Exercise 1.2

Can you verify the following equivalences related to Pauli matrices?

- $HXH = Z$
- $HYH = -Y$
- $HZH = X$

Exercise 1.3

Prove Theorem 1.5.1.

Exercise 1.4

Given $|u\rangle = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$, $|v\rangle = \begin{bmatrix} -2 \\ 1 \\ 3 \\ 0 \end{bmatrix}$,
compute $\langle u|v\rangle$ and $|u\rangle\langle v|$.

Capitolo 2

Lecture 4

This lecture covers the application of gates to qubits in a superposition state and explores the principles of unitary transformations in quantum mechanics, focusing on unitary matrices and their properties.

2.1 Applying an operation to a Superposition

In the previous lecture, we introduced some 2×2 matrix acting on a single qubit and demonstrated their actions on the computational basis states $|0\rangle$ and $|1\rangle$. When applied to a qubit in superposition, these matrices produce the following transformations:

$$\begin{aligned}\alpha |0\rangle + \beta |1\rangle &\xrightarrow{\sigma_x} \beta |0\rangle + \alpha |1\rangle, \\ \alpha |0\rangle + \beta |1\rangle &\xrightarrow{\sigma_y} -\beta i |0\rangle + \alpha i |1\rangle, \\ \alpha |0\rangle + \beta |1\rangle &\xrightarrow{\sigma_z} \alpha |0\rangle - \beta |1\rangle, \\ \alpha |0\rangle + \beta |1\rangle &\xrightarrow{H} \alpha |+\rangle + \beta |-\rangle, \\ \alpha |0\rangle + \beta |1\rangle &\xrightarrow{S} \alpha |0\rangle + \beta i |1\rangle, \\ \alpha |0\rangle + \beta |1\rangle &\xrightarrow{T} \alpha |0\rangle + e^{i\frac{\pi}{4}} \beta |1\rangle.\end{aligned}$$

2.2 Unitary Matrices

According to the principles of quantum mechanics, quantum systems evolve through linear operations which preserve the unit length of the quantum states, that is **unitary transformations**.

Definition 2 (Unitary Transformations) *A unitary transformation is a mapping that preserves the norm and inner product of vectors, ensuring the mapping of unit vectors to unit vectors.*

We are interested in some properties of these unitary transformations, and more particularly of unitary matrices that realize the unitary transformation. A d -dimensional quantum state space is a vector space generated, for example, by the vectors of the "computational basis" $\{|0\rangle, |1\rangle, \dots, |d-1\rangle\}$, where $|j\rangle$ is the vector with a 1 in the j -th position (0-based).

From the given definition of unitary transformation, we would like to better characterize a matrix $U = [c_0, \dots, c_{d-1}]$ representing a unitary map. We have $U|j\rangle = |c_j\rangle$, where c_j is a unit vector because the action of U should map unit vectors to unit vectors.

The properties of unitary transformations and unitary matrices include:

1. Columns of U are unit vectors, i.e. $\|c_j\|_2 = 1$, for all $j = 0, \dots, d-1$
2. U can be decomposed into a sum of rank-1 matrices:

$$U = \sum_{k=0}^{d-1} |c_k\rangle \langle k| = |c_0\rangle \langle 0| + |c_1\rangle \langle 1| + \dots + |c_{d-1}\rangle \langle d-1|.$$

Applying U to linear combinations of basis vectors yields:

$$U \left(\frac{|k\rangle + |j\rangle}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} (|c_k\rangle + |c_j\rangle).$$

Also, considering relative phases:

$$U \left(\frac{|k\rangle + i|j\rangle}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} (|c_k\rangle + i|c_j\rangle).$$

This ensures unit vector lengths, leading to $\langle c_j | c_k \rangle$ being real and purely imaginary, implying orthogonality: $\langle c_j | c_k \rangle = 0$. Hence,

3. Columns of U forms an orthonormal basis.

Other keys properties of unitary matrices include:

4. Rows of U are orthonormal as well.
5. Inverse of U is the conjugate transpose: $U^{-1} = U^\dagger$ (very important), where

$$(U^\dagger)_{ij} = \text{conj}(u_{ji}).$$

Different texts use different notations to denote the conjugate-transpose of a matrix $A^\dagger = A^H = A^*$. All these different symbols have the same meaning.

The spectral theorem for normal matrices states that a matrix is normal if and only if it can be diagonalized by a unitary transformation, a fundamental result in the study of unitary transformations and their relationship with eigenvectors and eigenvalues. In summary, unitary transformations and unitary matrices are crucial in quantum computing, allowing the manipulation of quantum states in a controlled and reversible manner.

2.2.1 Spectral Theorem

A key concept is the Spectral Theorem, which helps us understand how certain matrices can be diagonalized by means of unitary transformations.

For instance, symmetric matrices $A = A^T$ can be diagonalized by orthogonal matrices Q , such that $A = QDQ^T$, where D is a diagonal matrix containing real eigenvalues λ_i . Similarly, Hermitian matrices, which satisfy $A = A^\dagger$, can be diagonalized by unitary matrices. A more general case involves the so called **normal matrices**, which fulfill the condition $AA^\dagger = A^\dagger A$, and these matrices are diagonalizable by unitary transformations.

Theorem 2.2.1 (Spectral Theorem) *A matrix A is normal if and only if there exists a unitary matrix U and a diagonal matrix D such that $A = UDU^\dagger$. In this case, $Au_i = \lambda_i u_i$ for all $i = 1, \dots, n$, where u_i represents the columns of U , forming an orthonormal basis of eigenvectors, and λ_i are the corresponding eigenvalues.*

These properties allow us to express A as a sum of rank-1 matrices:

$$A = UDU^\dagger = \sum_{k=1}^n \lambda_k |u_k\rangle \langle u_k|,$$

where $U = [u_1 | \dots | u_n]$. Here, $P_k = |u_k\rangle \langle u_k|$ serves as an example of an orthogonal projector onto the eigenspace corresponding to λ_k .

2.2.2 Rotations

In quantum computing, rotations play a vital role. We can perform rotations of arbitrary angles θ around different axes, and these rotations are represented by unitary matrices. For example, a rotation around the x -axis is expressed as $R_x(\theta) = e^{-i\frac{\theta}{2}X}$ (similar rotations exist for the z and y axes). How is defined the exponential of a matrix M ?

If M is a normal matrix then we can use the Spectral Theorem 2.2.1 to decompose the matrix as $M = UDU^\dagger$, so we can define the exponential of a matrix as

$$e^M = U \left(\sum_{k=1}^{\infty} \frac{D^k}{k!} \right) U^\dagger = U e^D U^\dagger$$

For example, since $\sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is diagonal

$$R_z(\theta) = e^{-i\frac{\theta}{2}Z} = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix} = e^{-i\frac{\theta}{2}} \begin{bmatrix} 1 & 0 \\ 0 & -e^{i\theta} \end{bmatrix}.$$

To compute $e^{-i\theta/2 \sigma_k}$, $k \in \{x, y, z\}$:

1. Diagonalize σ_k

2. $U^\dagger \sigma_k U = D_k$
3. Compute $e^{-i\theta/2 D_k}$ which is diagonal
4. Set $e^{-i\theta/2 \sigma_k} = U e^{-i\theta/2 D_k} U^\dagger$.

In general, if M is one of the Pauli matrices, we have that

$$R_M(\theta) = e^{-i\theta/2 M} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} M, \quad (2.1)$$

which is derived observing that if $x \in \mathbb{R}$ and A is such that $A^2 = I$, which is true for the Pauli matrices, we have

$$e^{iAx} = \cos(x)I + i \sin(x)A.$$

Theorem 2.2.2 *For a 1-qubit unitary gate U , there exist real numbers $\alpha, \beta, \gamma, \delta$ such that U can be expressed as:*

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta).$$

This representation involves two rotations around the z -axis and one rotation around the y -axis, making it possible to decompose any unitary gate as a product of three rotations and a phase factor.

2.3 Exercises

Exercise 2.1

Show that there is no unitary U that can delete information, i.e. such that $U|\phi\rangle = |0\rangle$ for every 1-qubit quantum state $|\phi\rangle$.

Exercise 2.2

Prove that if you rotate of $\pi/2$ the $|+\rangle$ state you end up in state $|\bullet\rangle$. Verify on the Bloch sphere.

Exercise 2.3

Write the expression for $R_x(\theta)$ and $R_y(\theta)$ and verify that these rotations can be written as in equation 2.1.

Exercise 2.4

Prove the following identity

$$\langle u | A v \rangle = \langle A^\dagger u | v \rangle$$

Capitolo 3

Lecture 5

The **trace function**, denoted by $\text{tr}(A)$, is a fundamental concept in quantum computing. It calculates the sum of the diagonal elements of a matrix A , represented as $\sum_i A_{ii}$. The trace function exhibits linearity, satisfying $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ and $\text{tr}(\alpha A) = \alpha \text{tr}(A)$. Moreover, the *cyclic property* $\text{tr}(AB) = \text{tr}(BA)$ implies that the trace remains invariant under unitary transformations, i.e., $\text{tr}(UAU^\dagger) = \text{tr}(U^\dagger UA) = \text{tr}(A)$.

Another useful property is $\text{tr}(A|\phi\rangle\langle\phi|) = \langle\phi|A|\phi\rangle$, which is valuable in quantum computations.

3.1 Multi-Qubit Systems

Single qubits can be combined using the tensor product operation to create systems with multiple qubits. The dimension of a state space in quantum computing is always a power of two. The tensor product operation works as follows:

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}.$$

For example, in the case of $n = 2$ qubits, the state space consists of four basis states:

$$|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle.$$

Here, $|1\rangle \otimes |0\rangle$ denotes that the first qubit is in state $|1\rangle$ while the second qubit is in state $|0\rangle$. This state is often abbreviated as $|1\rangle|0\rangle = |10\rangle = |2\rangle$.

The dimension of the state space for n qubits is 2^n , and the basis states are of the form $|b_1\rangle \otimes |b_2\rangle \otimes \cdots \otimes |b_n\rangle$, where $b_i \in \{0, 1\}$. We can represent these basis states as $|0\rangle, |1\rangle, \dots, |2^n - 1\rangle$. Note that $|d\rangle$, with d being an integer between 0 and $2^n - 1$, is a vector of length 2^n with 1 in position d (starting from 0). The computational basis forms an orthonormal basis.

For matrices A and B , the tensor product is defined as:

$$A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{bmatrix}$$

The tensor product is not commutative, as illustrated by :

$$I \otimes \sigma_x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\sigma_x \otimes I = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

3.1.1 Entanglement and Degrees of Freedom

A system of n qubits (or a quantum register) can exist in a superposition, where the state is a linear combination of basis states. For instance, using the computational basis, we have:

$$\alpha_0 |0\rangle + \alpha_1 |1\rangle + \dots + \alpha_{2^n-1} |2^n - 1\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad \sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1.$$

With n qubits, the degrees of freedom are $2^{n+1} - 2$. This accounts for 2^n complex coefficients but with two constraints: normalization to one and the meaningless global phase factors. This leads to the classification of entangled and separable states.

Indeed, for example with $n = 2$ we have 6 degree of freedom, but if we multiply two qubits we get a state with only four degrees of freedom, meaning that there are the so-called entangled states.

Definition 3 *A state of a multi-qubit system is **entangled** when it cannot be expressed as the tensor product of its constituent qubits. Otherwise, it is called **separable**.*

The Bell state (EPR state) $\frac{|01\rangle + |10\rangle}{\sqrt{2}}$ is an example of an entangled state, while $(\alpha |0\rangle + \beta |1\rangle) \otimes (\delta |0\rangle + \gamma |1\rangle)$ represents a separable state.

Operations with tensor products In this section we give some interesting property of inner product applied to sequences of qubits. Consider two quantum separable states:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$$

$$|\phi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle$$

we have that their inner product is

$$\langle\psi|\phi\rangle = (\langle\phi_1| \otimes \langle\psi_2|)(|\phi_1\rangle \otimes |\phi_2\rangle) = \langle\psi_1|\phi_1\rangle \langle\psi_2|\phi_2\rangle. \quad (3.1)$$

One can simply verify on 2-qubits states.

Some useful properties include:

- (a) $c(|\psi_1\rangle_A \otimes |\psi_2\rangle_B) = c|\psi_1\rangle_A \otimes |\psi_2\rangle_B = |\psi_1\rangle_A \otimes c|\psi_2\rangle_B$
- (b) $(|\psi_1\rangle_A + |\psi_2\rangle_A) \otimes |\psi_3\rangle_B = |\psi_1\rangle_A \otimes |\psi_3\rangle_B + |\psi_2\rangle_A \otimes |\psi_3\rangle_B$
- (c) $(A \otimes B)(|\psi\rangle_A \otimes |\phi\rangle_B) = A|\psi\rangle_A \otimes B|\phi\rangle_B$
- (d) $(A \otimes B)(C \otimes D) = AC \otimes BD$
- (e) $(|\psi\rangle \otimes |\phi\rangle)^* = \langle\psi| \otimes \langle\phi|$
- (f) $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.

If you tensor-multiply a state by itself, we use the notation:

$$|\phi\rangle^{\otimes 3} = |\phi\rangle \otimes |\phi\rangle \otimes |\phi\rangle.$$

An important characteristic of quantum systems is the impossibility of copying an arbitrary quantum state.

Theorem 3.1.1 (No Cloning Theorem) *It is impossible to make a copy of an **unknown** quantum state.*

Proof: Since quantum operations are reversible, to produce two copies of a generic state $|\phi\rangle$, an additional ancillary qubit $|a\rangle$ is required (we can choose $|a\rangle = |0\rangle$).

Assume by contradiction that such that unitary U exists, then

$$U(|\phi\rangle \otimes |a\rangle) = |\phi\rangle \otimes |\phi\rangle. \quad (3.2)$$

If we apply U to another state $|\psi\rangle \otimes |a\rangle$, we would have back $|\psi\rangle \otimes |\psi\rangle$. Taking the inner product we have, observing that $U^H U = I$ and using (3.1)

$$(\langle\phi| \langle a|) U^H U (|\psi\rangle |a\rangle) = \langle\phi|\psi\rangle \underbrace{\langle a|a\rangle}_{=1} = \langle\phi|\psi\rangle$$

while, using equality 3.2 we get

$$(\langle\phi| \langle a|) U^H U (|\psi\rangle |a\rangle) = (\langle\psi|\phi\rangle)^2.$$

Hence, $\langle\psi|\phi\rangle = (\langle\phi|\psi\rangle)^2$ implying either $\langle\phi|\psi\rangle = 0$ or $\langle\phi|\psi\rangle = 1$. For the properties of the inner product, either $|\psi\rangle = |\phi\rangle$ or $|\phi\rangle$ is orthogonal to $|\psi\rangle$. Thus, we cannot copy a generic state.

3.2 Controlled operations

We also remind that for two unitary operators U_1 and U_2 , the operation $(U_1 \otimes U_2)(|\phi\rangle \otimes |\psi\rangle) = U_1 |\phi\rangle \otimes U_2 |\psi\rangle$.

We can represent this operation in a circuit:

$$\begin{array}{ccc} |\phi\rangle & \text{---} \boxed{U_1} \text{---} & U_1 |\phi\rangle \\ |\psi\rangle & \text{---} \boxed{U_2} \text{---} & U_2 |\psi\rangle. \end{array}$$

In quantum computing, gates that establish communication between different qubits are essential. Controlled operations, where an operation is performed only if a condition is met, are essential and widely used. One of the most common controlled operations is the C-NOT (controlled not or conditional-not) gate, which operates as follows:

In the classical framework, given two bits x and y , if $x = 0$ returns y ; if $x = 1$, return $\text{not}(y)$. Since in the quantum framework the computation should be reversible, we need a matrix acting on two qubits

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

We observe that C-NOT can copy classical information, it is sufficient to apply C-NOT to $|x0\rangle$

- C-NOT $|00\rangle = |00\rangle$
- C-NOT $|10\rangle = |11\rangle$
- C-NOT $|x0\rangle = |xx\rangle$

this is not violating the no-cloning theorem since x is a classical bit. However, when applied to a quantum state, the C-NOT gate can create entangled states.

For instance, applying C-NOT gate to $|\psi\rangle |0\rangle$ results in:

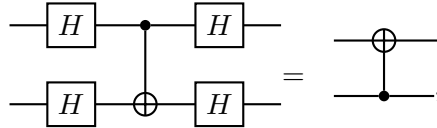
$$\text{CNOT } |\psi\rangle |0\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \\ 0 \\ \beta \end{bmatrix} = \alpha |00\rangle + \beta |11\rangle$$

meaning that C-NOT is able to create entangled states. An interesting case is when the target (the last one) qubit is in a superposition. In this case we can have the so-called **phase kickback**, i.e. the application of C-NOT kicks back to the control qubits in

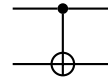
addition to the target qubit.

$$\text{CNOT} |+-\rangle = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = |--\rangle,$$

When the control is placed on the second qubit, we can establish the following equality:



where the circuital representation of the C-NOT gate is depicted as:

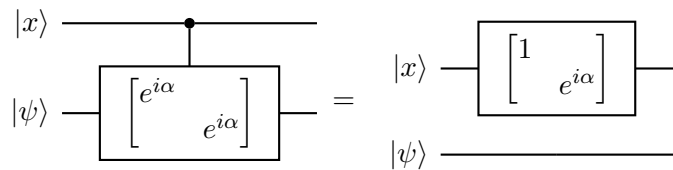


This circuital equivalence can be proven using matrices:

$$(H \otimes H)(\text{C-NOT})(H \otimes H) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

by verifying its action on a basis. This is another example of phase kickback.

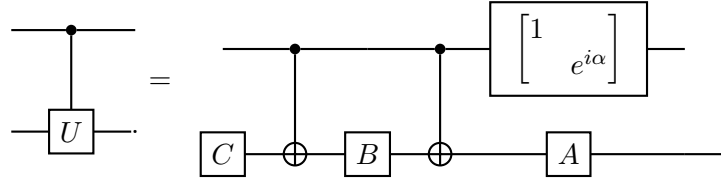
Kickback is where the eigenvalue added by a gate to a qubit is ‘kicked back’ into a different qubit via a controlled operation. For instance, applying an eigenvector to U yields no observable change due to the introduction of only a global phase factor, $U|\psi\rangle = e^{i\alpha}|\psi\rangle$. However, if we control this U using a qubit in superposition, the phase becomes relative rather than global, altering the relative phase in our control qubit:



It is interesting to explore how to implement a controlled- U operation for an arbitrary single-qubit gate U using only single-qubit operations and C-NOT gates.

Based on a generalization of Theorem 1.5.1, where $U = e^{i\alpha}AXBXC$ and $ABC = I$, we have:

Indeed we have,



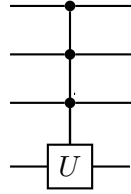
If the control qubit is $|1\rangle$, then the operation $U = e^{i\alpha}AXBXC$ is applied to the target qubit. Conversely, when the control is $|0\rangle$, nothing is done to the target qubit, consistent with $ABC = I$.

3.2.1 Multicontrolled operations

We can define also multi-controlled gates. Assume U is a k qubit unitary, controlled by n qubits. Then this operation acts on $n + k$ qubits and is defined as:

$$C^n(U) |x_1 x_2 \dots x_n\rangle |\psi\rangle = |x_1 x_2 \dots x_n\rangle U^{x_1 x_2 \dots x_n} |\psi\rangle,$$

where the exponent of U is the multiplication of the bits $x_1, x_2, \dots, x_n$ meaning that U is applied only if all the control qubits are 1. The circuitual representation is

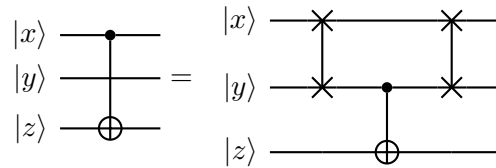


Another important 2 qubits quantum gate is the swap gate used to swap two qubits in the canonical basis.

We can represent this matrix using the swap gate which is realized by the following matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{array}{c} \text{---} \times \text{---} \\ | \\ \text{---} \times \text{---} \end{array}$$

We then have the following gate equality



you can recover with the matrices that

$$\begin{bmatrix} I & & & \\ & I & & \\ & & X & \\ & & & X \end{bmatrix} = (\text{SWAP} \otimes I)(I \otimes \text{C-NOT})(\text{SWAP} \otimes I).$$

Quantum System Evaluation In quantum computing, the state of a quantum system evolves based on the following postulates:

1. Isolated processes undergo unitary evolution: $|\psi_0\rangle \mapsto |\psi_1\rangle = U_0 |\psi_0\rangle \mapsto \dots |\psi_k\rangle = U_{k-1} |\psi_{k-1}\rangle$. This implies that a closed system evolves according to the Schrödinger equation:

$$i\hbar \frac{d|\psi\rangle}{dt} = H |\psi\rangle,$$

where H is a Hermitian operator known as the Hamiltonian of the system.

2. Periodically, the process is "observed" through experiments, a phenomenon called **measurement**.

Unitary matrices define valid quantum gates, with Pauli matrices and other similar matrices serving as one-qubit quantum gates. We call gates the unitary acting on a small number of qubits.

Exercise 3.1

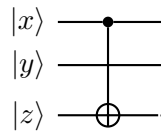
Prove the equality

$$H X H = Z.$$

Which are the eigenvalues of X .

Exercise 3.2

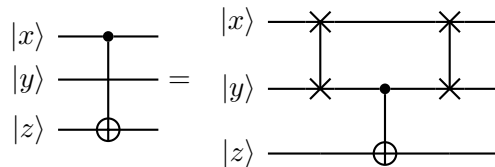
What is the effect of the following circuit?



Write down the matrix realizing the operation.

Exercise 3.3

Prove the following gate equality



Capitolo 4

Lecture 13

4.1 Grover's Algorithm

Grover's algorithm is a quantum algorithm used for searching in an unstructured set of elements, reducing the complexity from the classical $O(N)$ to $O(\sqrt{N})$. From the classical point of view this is a remarkable result since in the worst case scenario, we need to examine all the elements to find our target element if the dataset is unstructured. On the contrary, we can reduce the number of operations to $O(\log N)$ in the case the elements are sorted using the binary search algorithm. This result is due to Lev Grover in 1996.

Example: Diophantine equation Find $x, y, z, w \in \mathbb{N}$ with $x, y, z, w \leq 10^6$ such that $x^4 + y^4 + z^4 = w^4$. With a classical brute force approach we have a total number of queries of $\frac{10^{18}}{6}$ (removing permutations). With the quantum algorithm we can lower this number of queries to $\sqrt{N} = 10^9$.

If the solutions are more than one the algorithm, runs faster, in particular if you have M solutions (targets), you can lower the complexity to $O\left(\sqrt{\frac{N}{M}}\right)$.

4.1.1 Grover Iterations

For the moment assume $M = 1$, and that the target solution is denoted by β and that you are given a classical oracle that recognizes the solution that is

$$\text{Oracle}(x) = \begin{cases} 0 & x = \beta \\ 1 & x \neq \beta \end{cases}$$

We need to represent in the quantum world this oracle, for example in our example we need an oracle that checks if given $x, y, z \in \mathbb{N}$ the value $(x^4 + y^4 + z^4)^{1/4}$ is an integer. If this is the case, the Oracle returns 1 otherwise it returns 0. In any case we should undo the computation to get rid of the garbage and restore the initial values, because quantum computation should be reversible.

Assume to assign a label to each entry of our unsorted dataset. In this way our dataset is $\{|0\rangle, \dots, |N-1\rangle$ with the usual convention that we are representing states with $\log_2(N)$ qubits, and the Oracle is a boolean function which returns 1 only on β .

To realize with a unitary operator the oracle it is enough to mark with a negative phase the target state $|\beta\rangle$. The unitary associated to the oracle is then

$$U_\beta |x\rangle = \begin{cases} |x\rangle & x \neq \beta \\ -|x\rangle & x = \beta. \end{cases}$$

Looking at U_β as a matrix, this is the identity except for the entry (β, β) where it holds -1.

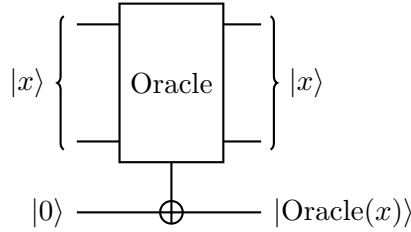
$$U_\beta = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & -1 & \\ & & & & \ddots \end{bmatrix} = \begin{bmatrix} (-1)^{\text{Oracle}(0)} & & & \\ & \ddots & & \\ & & \ddots & \\ & & & (-1)^{\text{Oracle}(N-1)} \end{bmatrix}$$

$$U_\beta |x\rangle = (-1)^{\text{Oracle}(x)} |x\rangle.$$

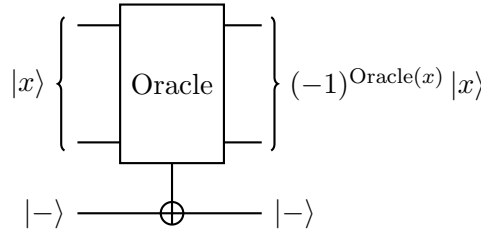
4.1.2 Construction of the Grover Oracle

It is always possible to convert a classical boolean function $f(x)$ to a reversible quantum circuit. The circuit is in xor with a $|0\rangle$ that emits $|f(x)\rangle$ such that if $f(x) = 0$ nothing happens, so $|0\rangle \rightarrow |0\rangle = |f(x)\rangle$.

If $f(x) = 1$ then $|0\rangle \rightarrow |1\rangle$.



In the case we initialize the ancillary qubit to $|-\rangle$ instead that to $|0\rangle$ we can change the phase only to the target phase while the ancilla register remains $|-\rangle$.



Indeed when $\text{Oracle}(x) = 0$ then the first and second register stay the same, however, when $\text{Oracle}(x) = 1$, we have a phase kickback and $|x\rangle |-\rangle \rightarrow |x\rangle |\text{Oracle}(x)\rangle = -|x\rangle |-\rangle = (-1)^{\text{Oracle}(x)} |x\rangle |-\rangle$.

Grover Algorithm

1. Initialize the circuit to a uniform superposition (usual Hadamard gates on each qubit), assuming $N = 2^n$

$$|\varphi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

2. Perform t times a **Grover Iteration**
3. Measure in the computational basis

The focus is on t : we need to estimate the number of iterations to have convergence to the target state.

Note that the application of the operator U_β changes only the relative phase, but we need another operation to amplify the amplitude of $|\beta\rangle$ because if we measure we have all the possible answers with the same probability $1/N$.

$$U_\beta |\varphi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0, x \neq \beta}^{N-1} |x\rangle - \frac{1}{\sqrt{N}} |\beta\rangle$$

So the Grover Iteration has the goal to amplify the amplitude of the target state in order to make it much bigger than the others. Two important states:

$|\varphi\rangle$ superposition of all the state

$|\beta\rangle$ the target state (good state)

Consider $\text{span}\{|\beta\rangle, |\varphi\rangle\}$ which is a plane of dimension two in $\mathbb{R}^N$. The two states $|\beta\rangle$ and $|\varphi\rangle$ are not orthogonal, indeed

$$\langle\beta|\varphi\rangle = \frac{1}{\sqrt{N}} \neq 0$$

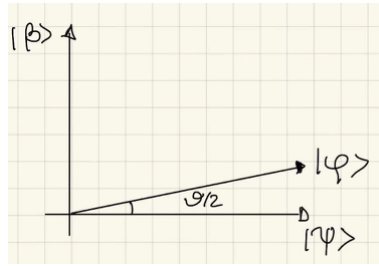
, but for very large N they are *almost orthogonal*. Consider a state $|\psi\rangle \in \text{span}\{|\beta\rangle, |\varphi\rangle\}$ such that $\langle\psi|\beta\rangle = 0$: we can do that taking $|\varphi\rangle$ and removing the components in directions of $|\beta\rangle$ and normalizing back.

$$|\psi\rangle = \sqrt{\frac{N}{N-1}} |\varphi\rangle - \frac{1}{\sqrt{N-1}} |\beta\rangle = \sqrt{\frac{N}{N-1}} \left(\frac{1}{\sqrt{N}} \sum_{x=0}^N |x\rangle \right) - \frac{1}{\sqrt{N-1}} |\beta\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \beta} |x\rangle.$$

We can now rewrite $|\varphi\rangle$ as a linear combination of the orthonormal states ψ and β , i.e.

$$|\varphi\rangle = \sin \frac{\theta}{2} |\beta\rangle + \cos \frac{\theta}{2} |\psi\rangle = \frac{1}{\sqrt{N}} |\beta\rangle + \sqrt{\frac{N-1}{N}} |\psi\rangle.$$

Indeed $\langle\varphi|\psi\rangle = \cos \frac{\theta}{2}$, and $\langle\varphi|\beta\rangle = \sin \frac{\theta}{2}$. So the angle between $|\psi\rangle$ and $|\varphi\rangle$ is $\frac{\theta}{2}$, and $\sin \frac{\theta}{2} = 1/\sqrt{N}$.

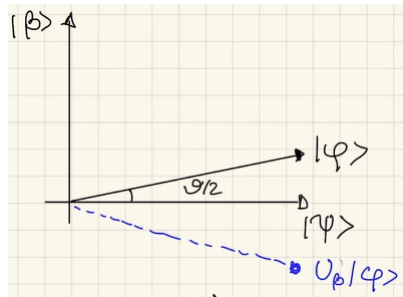


ψ and ϕ are inverted

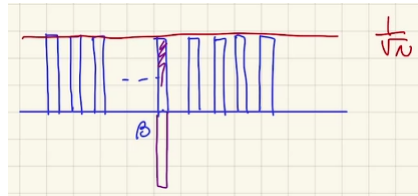
So we start with $|\varphi\rangle$ and apply U_β to it, the effect is to mark β changing the phase of entry β .

$$U_\beta |\varphi\rangle = \cos \frac{\theta}{2} |\phi\rangle - \sin \frac{\theta}{2} |\beta\rangle$$

This is a reflection along $|\psi\rangle$, the space orthogonal to $|\beta\rangle$.



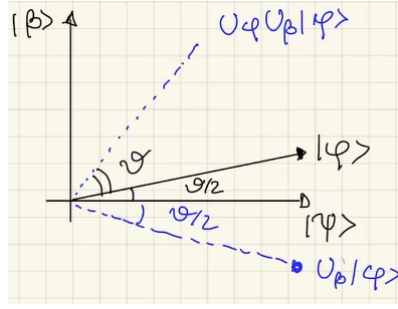
Starting with amplitudes all equal to $\frac{1}{\sqrt{N}}$, the amplitude of $|\beta\rangle$ is inverted.



Then we can apply a diffuser step to amplify the amplitude of β . This can be done with U_φ which reflects around $|\varphi\rangle$. In particular

$$U_\varphi = 2|\varphi\rangle\langle\varphi| - I$$

We obtain $U_\varphi U_\beta |\varphi\rangle$ which reflects $U_\beta |\varphi\rangle$ around $|\varphi\rangle$ with an angle of θ with $|\varphi\rangle$



This reflection is often referred to as inversion about the mean. Let us denote by G the composition of the two reflections:

$$G = U_\varphi U_\beta,$$

and, starting with $|\varphi^{(0)}\rangle = |\varphi\rangle$ define the sequence $|\varphi^{(k+1)}\rangle = G|\varphi^{(k)}\rangle$.

If $|z\rangle = U_\beta \varphi^{(k)} = \sum_{x=0}^{N-1} \alpha_x |x\rangle$. When $k = 0$ $\alpha_x = \frac{1}{\sqrt{N}}$ if $x \neq \beta$ and $\alpha_x = -\frac{1}{\sqrt{N}}$ if $x = \beta$. Applying the diffuser we have

$$|\varphi^{(k+1)}\rangle = U_\varphi |z\rangle = (2|\varphi\rangle\langle\varphi| - I)|z\rangle = 2 \sum_{x=0}^{N-1} \alpha_x \langle\varphi|x\rangle |\varphi\rangle - \sum_{x=0}^{N-1} \alpha_x |x\rangle.$$

We have $\langle\varphi|x\rangle = \frac{1}{\sqrt{N}}$ while $|\varphi\rangle$ does not depend on $|x\rangle$ and can be brought outside the summation,

$$|\varphi^{(k+1)}\rangle = |\varphi\rangle \left(2 \sum_{x=0}^{N-1} \frac{1}{\sqrt{N}} \alpha_x \right) - |z\rangle.$$

Moreover we have

$$2 \sum_{x=0}^{N-1} \frac{1}{\sqrt{N}} \alpha_x = \frac{2N}{\sqrt{N}} \frac{1}{N} \sum_{x=0}^{N-1} \alpha_x = 2\sqrt{N} M,$$

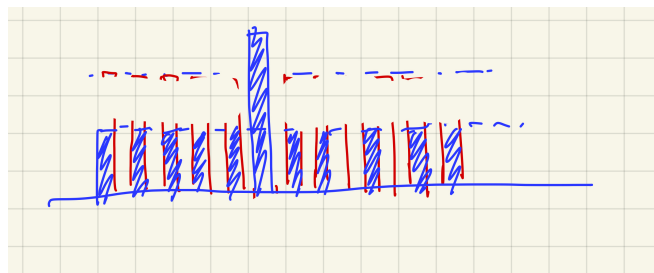
where $M = \frac{1}{N} \sum_{x=0}^{N-1} \alpha_x$, is the average of the values in z . We can rewrite it as

$$|\varphi^{(k+1)}\rangle = \sum_{x=0}^{N-1} (-\alpha_x + 2M) |x\rangle.$$

The first step

$$|\psi^{(1)}\rangle = \left(2M - \frac{1}{\sqrt{N}} \right) \sum_{x \neq \beta} |x\rangle + \left(2M + \frac{1}{\sqrt{N}} \right) |\beta\rangle$$

Can see that the amplitude of $|\beta\rangle$ has increased and the others have decreased by a factor proportional to the mean.



How many iterations do we need to estimate accurately $|\beta\rangle$? $G = U_\varphi U_\beta$ is a rotation of an angle θ . Indeed the composition of two reflections is always a rotation! Remember that θ is such that $\sin \frac{\theta}{2} = \frac{1}{\sqrt{N}}$ hence, $\theta = 2 \arcsin \frac{1}{\sqrt{N}}$.

Globally, G^k performs a rotation of $k\theta$ respect to $|\varphi\rangle$. We stop when $k\theta + \frac{\theta}{2} = \frac{\pi}{2}$ (near $|\beta\rangle$). This means $k = (\frac{\pi}{2} - \frac{\theta}{2}) \frac{1}{\theta}$. But $\frac{\pi}{2}$ is irrational, and k should be an integer, so we want an approximation $(2k+1)\frac{\theta}{2} \simeq \frac{\pi}{2}$

$$k = \left\lceil \frac{\pi}{2\theta} - \frac{1}{2} \right\rceil,$$

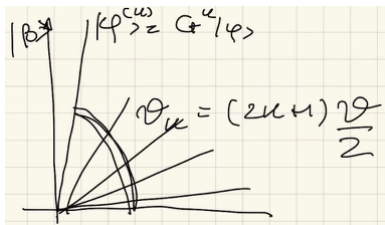
i.e. k is the closest integer to $\frac{\pi}{2\theta} - \frac{1}{2}$.

For a sufficiently large N we can approximate $\arcsin\left(\frac{1}{\sqrt{N}}\right) = \frac{1}{\sqrt{N}} = \frac{\theta}{2}$ (from the Taylor expansion of $\arcsin$) and we get

$$k = \left\lceil \left(\frac{\pi}{2} - \frac{1}{\sqrt{N}} \right) \frac{\sqrt{N}}{2} \right\rceil = \left\lceil \frac{\pi}{4} \sqrt{N} - \frac{1}{2} \right\rceil$$

So the number of iterations is

$$k \simeq \frac{\pi}{4} \sqrt{N}$$



and the angle between $|\beta\rangle$ and $|\varphi^{(k)}\rangle$ is $\frac{\pi}{2} - (2\frac{\pi}{4}\sqrt{N} + 1)\frac{\theta}{2} = \frac{\pi}{2} - (\frac{\pi}{4}\sqrt{N}\theta + \frac{\theta}{2})$

Theorem 4.1.1 For large N , if we perform $k = \left\lceil \frac{\pi}{4} \sqrt{N} \right\rceil$ Grover Iterations, the probability of measuring the target state $|\beta\rangle$ is

$$P(\text{Measuring } |\beta\rangle) = 1 - \frac{1}{N}.$$

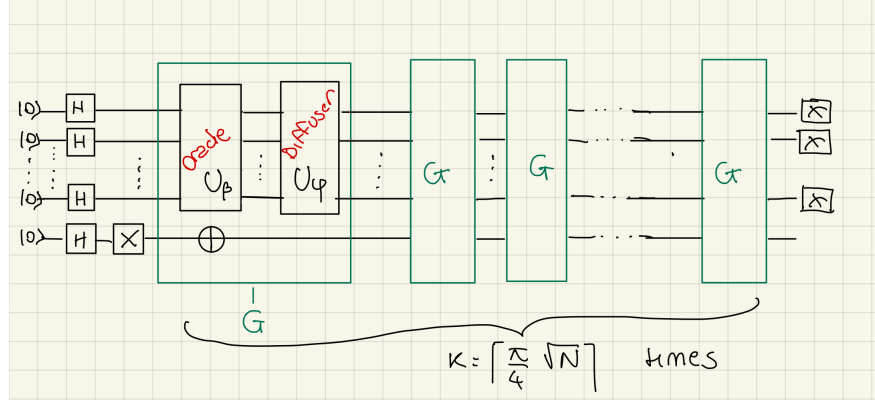
Proof. Projecting along $|\beta\rangle$ at time k

$$P(\text{Measuring } |\beta\rangle) = |\langle \beta | G^k \varphi \rangle|^2 = \cos^2 \left(\frac{\pi}{2} - (2k+1)\frac{\theta}{2} \right) = \cos^2 \left(\frac{\theta}{2} \right) = 1 - \sin^2 \left(\frac{\theta}{2} \right) = 1 - \frac{1}{N}.$$

4.1.3 Circuit for the Grover Iteration G

We can create the equal superposition $|\varphi\rangle$ applying an Hadamard gate to each input qubit, $|\psi\rangle = H^{\otimes n} |0\rangle$. The oracle U_β is implemented as described before, and depends on the boolean function realizing the Oracle. To implement the diffuser $U_\varphi = 2|\varphi\rangle\langle\varphi| - I$ Note that $U_\varphi = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$.

The Grover algorithm is schemitized as follows



1. Initialize the registers to $|0\rangle^{\otimes n} |0\rangle$, $N = 2^n$.

2. Apply an Hadamard gate to each of the $n + 1$ qubit and a X gate to the last qubit. We obtain $|\varphi\rangle |-\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$ is obtained applying H to each qubit and an X gate on the last one

3. Apply $k = \left\lceil \frac{\pi}{4} \sqrt{2^n} \right\rceil$, operator $G = U_\varphi U_\beta$ to the first n qubits and the identity to the last one. We end up with

$$(U_\varphi U_\beta \otimes I)^k |\varphi\rangle |-\rangle \approx |\beta\rangle |-\rangle.$$

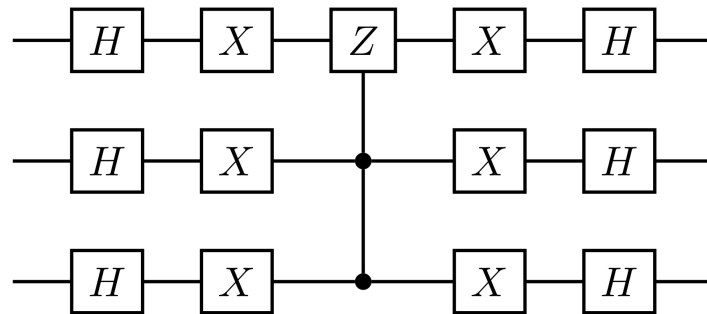
4. Measure the first n qubits to get β .

Exercise 4.1

Show that applying U_φ to a general state $\sum_x \alpha_x |x\rangle$ produces $\sum_x (-\alpha_x + 2M) |x\rangle$, where $M = \frac{1}{N} \sum_x \alpha_x$ is the mean value of the α_k .

Exercise 4.2

Verify that the following circuit realizes the diffuser operator.



Capitolo 5

Lecture 17

5.1 The Quantum Fourier Transform

The Quantum Fourier transform is for quantum computing as important as its classical counterpart, the Fast Fourier transform is for classical computation. Indeed is the building block for many important algorithms: QPE (Quantum Phase Estimation), QAA (Quantum Amplitude Amplification), Shor's Algorithm, HHL (for the solution of linear systems)

5.1.1 The Discrete Fourier Transform

Before presenting the QFT we need to describe the Discrete Fourier Transform (DFT) which consists essentially of multiplying a particular unitary matrix - the Fourier matrix - by a vector. This operation has several important applications ranging from application in signal and image processing, data compression, and computation of the solution of partial differential equations, just to mention some. Fast Fourier Transform is the algorithm that allows to compute the DFT of a vector with N entries in $O(N \log N)$ with $N = 2^n$ instead of the $O(N^2)$ operations needed to perform matrix-vector multiplication.

Let $x = [x_0, \dots, x_{N-1}]$ and we compute $y = [y_0, \dots, y_{N-1}] = DFT(x)$ as follows

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{\frac{2\pi i j k}{N}} x_j = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega_N^{jk} x_j,$$

where $\omega_N = e^{\frac{2\pi i}{N}}$ are the primitive N -th roots of unity, i being the imaginary unit.

Vector y corresponds to the unitary transformation of x by means of the Fourier matrix F_N , defined as

$$F_N = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & \dots & \dots & 1 \\ 1 & \omega_N & \dots & \omega_N^{N-1} \\ \vdots & & & \\ 1 & \omega_N^{N-1} & \dots & \omega_N^{(N-1)(N-1)} \end{bmatrix},$$

that is

$$(F_N)_{kj} = \omega_N^{kj} \text{ for } k, j = 0, \dots, N-1.$$

To prove that F_N is indeed an unitary matrix we use the fact that $\text{conj}(\omega_N^{jk}) = \omega_N^{-jk}$ because $\text{conj}(\omega_N^{jk}) = \text{conj}(\cos(2\pi jk/N) + i \sin(2\pi jk/N)) = \cos(2\pi jk/N) - i \sin(2\pi jk/N) = \cos(-2\pi jk/N) + i \sin(-2\pi jk/N) = \omega_N^{-jk}$. Using this fact and using the following result, one can prove that F_N is unitary.

Theorem 5.1.1

$$\sum_{j=0}^{N-1} \omega_N^{kj} = \begin{cases} N & k \equiv 0 \text{ mod } N \\ 0 & k \not\equiv 0 \text{ mod } N \end{cases}$$

Proof 1 *Indeed, if $k \equiv 0 \text{ mod } N$ then $k = sN$ and $\omega_N^{sNj} = (\omega_N^N)^{sj} = 1$ and hence $\sum_j = 0^{N-1} 1 = N$. If $k \not\equiv 0 \text{ mod } N$ then $\omega_N^k \neq 1$. If we set $\omega_N^k = x$, from the equality $(1 + x + x^2 + \dots + x^{N-1}) = 1 - x^N$, we have the thesis observing that $x^N = \omega_N^{Nk} = 1$.*

Indeed

$$(F_N F_N^H)_{st} = \frac{1}{N} \sum_{j=0}^{N-1} \omega_N^{js} \omega_N^{-jt} = \frac{1}{N} \sum_{j=0}^{N-1} \omega_N^{j(s-t)} = \begin{cases} 1 & s = t \\ 0 & s \neq t. \end{cases} \text{ cc}$$

Example 1 *For $N = 2$, we have that*

$$F_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H.$$

Note that $|y\rangle = H^{\otimes n} |x\rangle$ it's already a QFT over the group $\mathbb{Z}_2 \otimes \dots \otimes \mathbb{Z}_2$.

5.2 Quantum Fourier Transform

Coppersmith (1994) developed the Quantum Fourier Transform (QFT), which transitions from the $O(N \log N)$ complexity of the classical Fast Fourier Transform to $O(n^2)$ in the quantum realm, achieving an exponential speedup.

the QFT operates differently from its classical counterpart. It computes a state vector whose amplitudes correspond to the scaled entries of $y = DFT(x)$. However, when the quantum state is measured, the superposition is destroyed, and we obtain a binary sequence. In the classical case, we retrieve all the components of $Y = DFT(x)$, while in the quantum case, we obtain only a specific configuration after measurement. To approximate the amplitudes of the state vector corresponding to frequencies, the quantum circuit must be executed multiple times. This is why the QFT is often used as a subroutine within other quantum algorithms.

We can describe the algorithm on the computational basis $|0\rangle, |1\rangle, \dots, |N-1\rangle$ and then for linearity apply it to any state $|x\rangle$. We denote with the symbol $\tilde{\cdot}$ the vector transformed with the QFT

$$|\tilde{j}\rangle = QFT(|j\rangle)$$

More in general, with

$$|x\rangle = \sum_{j=0}^{N-1} x_j |j\rangle$$

$$|\tilde{x}\rangle = QFT(|x\rangle) = \sum_{j=0}^{N-1} x_j QFT(|j\rangle) = \sum_{j=0}^{N-1} x_j |\tilde{j}\rangle.$$

To define the QFT, we start from the classical definition applying to each element of the basis

$$|\tilde{j}\rangle = QFT(|j\rangle) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle$$

So QFT transforms from the computational basis to the Fourier basis.

Example 2 $n = 1$, 1-qubit system, $N = 2^1 = 2$, from the computational basis $\{|0\rangle, |1\rangle\}$ to the Fourier basis $\{|+\rangle, |-\rangle\}$.

$$|\tilde{0}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad |\tilde{1}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

indeed $|\tilde{0}\rangle = |+\rangle$ and $|\tilde{1}\rangle = |-\rangle$. We can represent either states as

$$|\tilde{j}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{\pi i j} |1\rangle). \quad (5.1)$$

5.2.1 Systems of n -qubits

With n -qubits we have the basis $\{|0\rangle, |1\rangle, \dots, |N-1\rangle\}$, $N = 2^n$, keeping in mind that it's a short-hand writing of a the binary string representing the integer with n bits. We have 2^n basis states so 2^n configuration of bits.

E.g. $|4\rangle = |100\rangle$ for $N = 2^3$. For $n = 3$

$$|\tilde{j}\rangle = \frac{1}{\sqrt{2^3}} \sum_{k=0}^{2^3-1} e^{\frac{2\pi i j k}{2^3}} |k\rangle,$$

Remembering that $|k\rangle = |k_1 k_2 k_3\rangle$ and that the sum must be interpreted as

$$\sum_k = \sum_{k_1=0}^1 \sum_{k_2=0}^1 \sum_{k_3=0}^1$$

In the general case

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{2^n}} |k\rangle,$$

with $k = k_1 k_2 \dots k_n$, the digits of the binary representation, such that $k = \sum_{s=1}^n k_s 2^{n-s}$ we have

$$\begin{aligned} |\tilde{j}\rangle &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j \sum_{s=1}^n k_s 2^{-s}} |k_1 \dots k_n\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \prod_{s=1}^n e^{\frac{2\pi i j k_s}{2^s}} |k_1 \dots k_n\rangle. \end{aligned}$$

Isolating in turn each qubit, we end up with

$$\begin{aligned} |\tilde{j}\rangle &= \\ &= \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2^1}} |1\rangle \right) \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^2}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) \\ &= \bigotimes_{s=1}^n \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j}{2^s}} |1\rangle \right). \end{aligned} \quad (5.2)$$

Each qubit has been transformed into something very similar to an Hadamard gate, except for the amplitude of $|1\rangle$.

So we went from

$$|j\rangle = |j_1 \dots j_n\rangle = |j_1\rangle \otimes \dots \otimes |j_n\rangle e$$

to

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2^1}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right)$$

with a one-to-one correspondence between qubits,

$$|j_s\rangle \mapsto |\tilde{j}_s\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j}{2^s}} |1\rangle \right).$$

Note that QFT does not create entanglement since the state in output is separable.

We now analyze how we can transform qubit j_s into $\frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j}{2^s}} |1\rangle \right)$.

Observe that we gets a bit of information from each configuration, with s affecting the relative phase.

$$\frac{1}{\sqrt{N}} e^{\frac{2\pi i j}{2}} \rightarrow |10 \dots 0\rangle$$

$$\frac{1}{\sqrt{N}} e^{\frac{2\pi i j}{2^2}} \rightarrow |0 \dots 01\rangle$$

$$\frac{1}{\sqrt{N}} e^{\frac{2\pi i j}{2^s}} \rightarrow |0 \dots 010 \dots 0\rangle \text{ where the 1 is in position } s$$

$$\frac{1}{\sqrt{N}} \left(e^{\frac{2\pi i j}{2}} \dots e^{\frac{2\pi i j}{2^n}} \right) = \frac{1}{\sqrt{N}} e^{2\pi i (\frac{j}{2} + \dots + \frac{j}{2^n})} \rightarrow |1 \dots 1\rangle.$$

So the phase applied is qubit-dependent.

Note moreover that $\frac{j}{2^s} = j_1 j_2 \dots j_{n-s} j_{n-s+1} \dots j_n$, meaning that the $n - s$ most significant digits of j do not enter into the calculation of this phase factor, since $e^{\frac{2\pi i j}{2^s}} = e^{2\pi i (j_1 j_2 \dots j_{n-s})} e^{2\pi i (0.j_{n-s+1} \dots j_n)} = 1 \cdot e^{2\pi i (0.j_{n-s+1} \dots j_n)}$.

Example 3 $n = 3, |j\rangle = |101\rangle$

$$|\tilde{5}\rangle = \frac{1}{N} \left(|0\rangle + e^{2\pi i \frac{5}{2}} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i \frac{5}{4}} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i \frac{5}{8}} |1\rangle \right)$$

5.2.2 The Circuit

As we remarked in (5.1)

$$H |j_s\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i}{2} j_s} |1\rangle \right).$$

A second ingredient is a rotation depending on index s such that

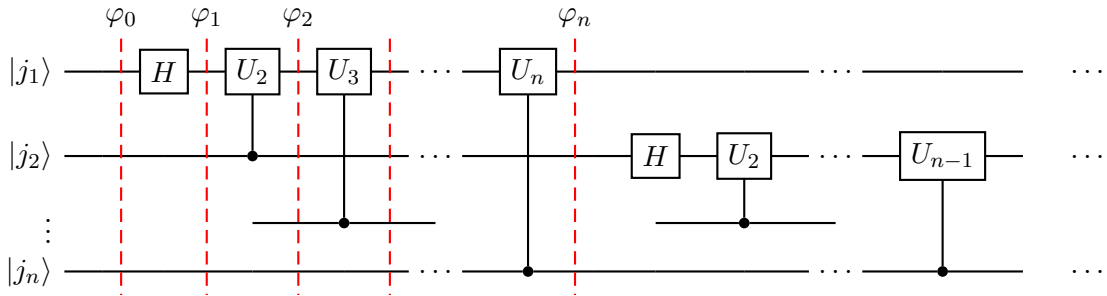
$$U_s |j_t\rangle = e^{\frac{2\pi i}{2^s} j_t} |j_t\rangle,$$

i.e. U_s just apply a phase to $|j_t\rangle$. If $j_t = 0$ then $e^{\frac{2\pi i}{2^s} j_t} |j_t\rangle = |0\rangle$, if $j_t = 1$ then $e^{\frac{2\pi i}{2^s} j_t} |j_t\rangle = e^{\frac{2\pi i}{2^s}} |1\rangle$ so we have

$$U_s = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^s}} \end{bmatrix}.$$

5.2.3 The Circuit

We will need controlled gates because we apply the phase only to particular configurations of qubits.



1. The initial state is $|\varphi_0\rangle = |j_1 \dots j_n\rangle$
2. $|\varphi_1\rangle = H |j_1\rangle \otimes |j_2 \dots j_n\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i}{2} j_1} |1\rangle \right) \otimes |j_2 \dots j_n\rangle$

3. Apply the gate U_s to qubit j_1 only when $j_2 = 1$, we have

$$|\varphi_2\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i}{2} j_1} e^{\frac{2\pi i}{2^2} j_2} |1\rangle \right) \otimes |j_2 \dots j_n\rangle$$

4.

$$|\varphi_3\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i}{2^3} j_3} e^{\frac{2\pi i}{2^2} j_2} e^{\frac{2\pi i}{2^1} j_1} |1\rangle \right) \otimes |j_2 \dots j_n\rangle$$

$\vdots$

n .

$$\begin{aligned} |\varphi_n\rangle &= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \left(\frac{j_n}{2^n} + \frac{j_{n-1}}{2^{n-1}} + \dots + \frac{j_1}{2^1} \right)} |1\rangle \right) \otimes |j_2 \dots j_n\rangle \\ &= \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i \left(\frac{j_1}{2^1} + \frac{j_2}{2^2} + \dots + \frac{j_n}{2^n} \right)} |1\rangle \right) \otimes |j_2 \dots j_n\rangle \end{aligned}$$

And given that $j = \sum_{s=1}^n j_s 2^{n-s} = 2^n \sum_{s=1}^n j_s 2^{-s}$ we have that $\frac{j}{2^n} = \sum_{s=1}^n \frac{j_s}{2^s}$ hence

$$|\varphi_n\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) \otimes |j_2 \dots j_n\rangle,$$

we have transformed the first qubit.

Apply the same reasoning we can transform all the other qubits. At the end we have

$$\frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^{n-1}}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^1}} |1\rangle \right).$$

So, comparing with (5.2) we see that the circuit implements QFT in the reverse order. We recover the desired configuration by applying for example swap gates at the beginning or the end of the circuit.

We see that there are n gates acting on each of the n qubits so the circuit uses at most n^2 controlled gates. This represents an algorithm with an exponential advantage over the classical counterpart.

We can observe moreover that for large s the rotation U_s is close to the identity matrix, since $e^{2\pi i/2^s} \approx 1$, for large s . In practice we can omit the application of U_s for $s > \log n$ and the cost decreases to $O(n \log n)$.

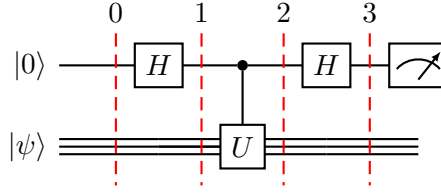
We note that the application of QFT to a generic state $|x\rangle$ which is the superposition of the basis states of the computational base $|x\rangle = \sum_{j=0}^{N-1} x_j |j\rangle$ is not accessible by measuring. Indeed $|\tilde{x}\rangle = QFT(|x\rangle) = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \sum_{k=0}^{N-1} x_j e^{\frac{2\pi i j k}{N}} |k\rangle$, so the action of measuring collapses the system in a computational basis state with probability proportional to the square of the amplitudes $|x_j|^2$. The execution of the reversed circuit gives us the inverse QFT.

5.3 Quantum Phase Estimation

Any unitary matrix is diagonalizable and has an orthonormal basis of eigenvectors. The eigenvalues are all in the form $e^{2\pi i\theta}$. Denoting the associated eigenstate (eigenvector) as $|\psi\rangle$, i.e.

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle,$$

the **QPE problem** is formulated as follows. If we can prepare $|\psi\rangle$ and we can apply U , can we extract θ , the phase that is the angle of the rotation? Let's see the intuition on how we could do without QFT.



0. The initial state is $|0\rangle|\psi\rangle$
1. $|\psi\rangle$ does not change and we apply H to $|0\rangle$ which becomes $|+\rangle$
The state is

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle|\psi\rangle + \frac{1}{\sqrt{2}}|1\rangle|\psi\rangle$$

2. U is applied only on the second qubit when the first is $|1\rangle$.
The state is

$$\frac{1}{\sqrt{2}}|0\rangle|\psi\rangle + \frac{1}{\sqrt{2}}e^{2\pi i\theta}|1\rangle|\psi\rangle$$

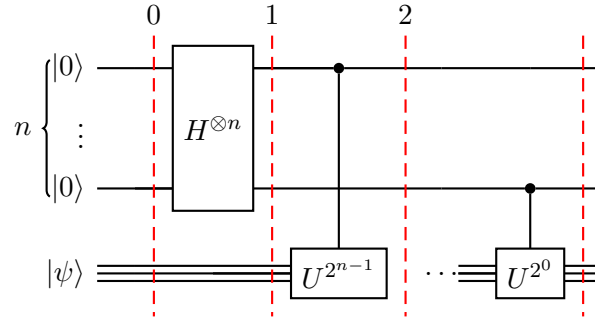
3. $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$, and H is applied only on the first qubit.

The state is

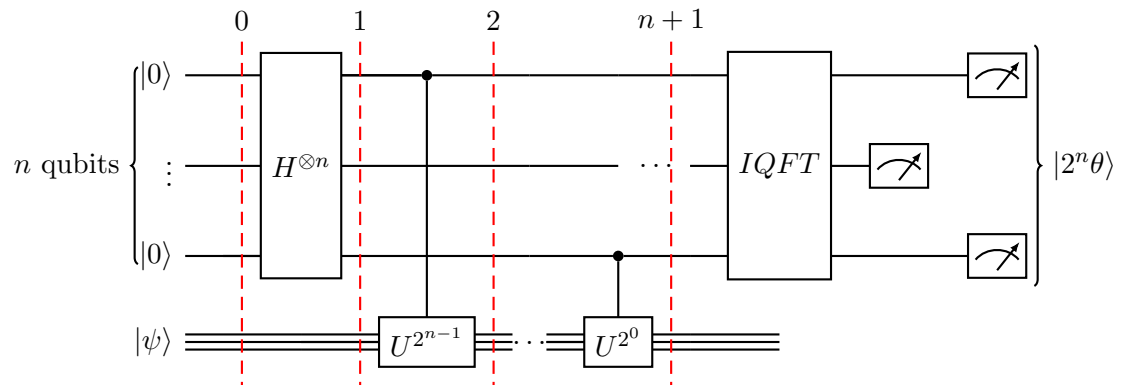
$$\begin{aligned} & \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)|\psi\rangle + \frac{1}{\sqrt{2}}e^{2\pi i\theta}\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)|\psi\rangle = \\ & = \frac{1}{2}\left(\left(1 + e^{2\pi i\theta}\right)|0\rangle + \left(1 - e^{2\pi i\theta}\right)|1\rangle\right)|\psi\rangle. \end{aligned}$$

The probability of measuring $|0\rangle$ in the first qubit is determined by the square of its corresponding amplitude. Since the amplitudes of $|0\rangle$ and $|1\rangle$ are different, we can rely on repeated experiments to estimate the amplitudes by measuring the frequencies of different outcomes over multiple trials. From the estimated amplitudes one can then recover the phase θ . From the Chebyshev inequality (or the Chernoff bound) we see that we need $1/\varepsilon^2$ repetition of the experiment for approximating the amplitudes with precision ε .

We can however estimate $\theta \in \mathbb{R}$ using the QFT with a lower complexity, the algorithm QPE returns the binary-encoding of θ on n -bit. For the QPE circuit, we need to decide the number n which is the number of bits for the binary encoding of the angle θ .



We first put in a superposition the n qubits and then we apply powers of U controlled by these ancillary qubits. Finally an IQFT transform is applied. The circuit becomes



Let us prove the correctness of this approach.

0. $|\varphi_0\rangle = |0\rangle^{\otimes n} |\psi\rangle$;
1. $|\varphi_1\rangle = \left(\frac{1}{\sqrt{2}}\right)^n (|0\rangle + |1\rangle)^{\otimes n} |\psi\rangle$;

Since $U^k |\psi\rangle = e^{2\pi i \theta k} |\psi\rangle$, and the control qubit is in superposition, so by applying the U gate the phase kicks back to the control qubit.

$$\begin{bmatrix} I & \\ & U^k \end{bmatrix} \left(\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \otimes |\psi\rangle \right) = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} e^{2\pi i \theta k} \end{bmatrix} \otimes |\psi\rangle$$

$n+1$.

$$\begin{aligned} |\varphi_{n+1}\rangle &= \left(\frac{1}{\sqrt{2}}\right)^n (|0\rangle + e^{2\pi i \theta 2^{n-1}} |1\rangle) \otimes (|0\rangle + e^{2\pi i \theta 2^{n-2}} |1\rangle) \otimes \dots \otimes (|0\rangle + e^{2\pi i \theta 2^0} |1\rangle) \otimes |\psi\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i \theta k} |k\rangle. \end{aligned}$$

Now to retrieve θ , we note that $|\varphi_{n+1}\rangle$ is the QFT of the binary encode of $\lceil 2^n \theta \rceil$. Indeed, the QFT of $|j\rangle$ is

$$|\tilde{j}\rangle = \frac{1}{\sqrt{N}} \left(|0\rangle + e^{\frac{2\pi i j}{2}} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\frac{2\pi i j}{2^n}} |1\rangle \right) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{2^n}} |k\rangle.$$

These two configurations are very similar, they differ for the exponent. So it's the same of QPE except that θ is replaced by $\frac{j}{2^n}$. Note that

$$IQFT(|k\rangle) = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{-2\pi i k \frac{j}{2^n}} |j\rangle,$$

hence, applying IQFT to state $|\varphi_{n+1}\rangle$, for the linearity of the operator we have

$$\begin{aligned} IQFT\left(\frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i \theta k} |k\rangle\right) &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i \theta k} IQFT(|k\rangle) \\ &= \sum_{j=0}^{2^n-1} \left(\frac{1}{2^n} \sum_{k=0}^{2^n-1} e^{2\pi i k(\theta - \frac{j}{2^n})} \right) |j\rangle. \end{aligned}$$

Since there exists an index $j' \in \{0, 1, \dots, 2^n - 1\}$ such that $\theta = \frac{j'}{2^n}$ we have that $\frac{1}{2^n} \sum_{k=0}^{2^n-1} e^{2\pi i k(\theta - \frac{j'}{2^n})} = 1$, that is the amplitude of the configuration j' is one and when we measure the ancilla qubits we end up with certainty in the configuration $j' = 2^n \theta$.

Capitolo 6

Lecture 21

For this lecture refer also to *Lecture Notes on Quantum Algorithms for Scientific Computing* by Lin Lin.

6.1 HHL: an Algorithm for the Solution of Linear systems

We consider a linear system

$$Ax = b,$$

where $A \in \mathbb{C}^{N \times N}$ is a non-singular matrix. We will assume also that A is hermitian, i.e. $A = A^\dagger$, otherwise we consider the matrix of size $2N$

$$\tilde{A} = \begin{bmatrix} 0 & A^\dagger \\ A & 0 \end{bmatrix}$$

and the enlarged system $\tilde{A} \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix}$.

Since the vector $b \in \mathbb{C}^n$ should be stored in a quantum state we need also b to be normalized to 1. The quantum algorithm will return a quantum state $|x\rangle$ which is normalized as well. This means that $|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}$, and we need a strategy to recover the normalizing constant.

QLSP problem: Find a quantum state $|\tilde{x}\rangle$ such that $\| |\tilde{x}\rangle - |x\rangle \| \leq \varepsilon$ where $|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}$.

We anticipate that the quantum algorithm will not allow the computation of all the entries of $|\tilde{x}\rangle$ but only some global properties of particular entries. In fact, the quantum state is in superposition and we will need to estimate the amplitudes of the state vector running many times the algorithm to recover the histogram.

6.1.1 Classical complexity, and some ideas for the quantum algorithm

Direct method for the solution of a dense linear system requires $O(N^3)$ floating point operations. However if the system is sparse, with an average of s non-zero entries per

row, we can employ Conjugate gradient methods or other Krylov methods which allows to solve the system in $O(Ns\mu(A)\log(\frac{1}{\varepsilon}))$ operation or even $O(Ns\sqrt{\mu(A)}\log(\frac{1}{\varepsilon}))$ in the case A is positive definite.

The idea behind the HHL algorithm is very simple. Since A is hermitian we know that there is a full set of orthonormal vectors $|v_j\rangle$ such that

$$A|v_j\rangle = \lambda_j|v_j\rangle, \quad j = 0, \dots, N-1.$$

If we know all the eigenvectors and the eigenvalues then we can obtain the inverse of A as

$$A^{-1} = VD^{-1}V^\dagger,$$

since $A = VDV^\dagger$, $V = [|v_0\rangle |v_1\rangle \dots |v_{N-1}\rangle]$ is the unitary matrix whose columns are the eigenvectors, and $D = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_{N-1})$.

In this way we can compute $x = A^{-1}b = VD^{-1}V^\dagger b$.

Since the columns of V are a basis of $\mathbb{C}^n$, we can express b as a linear combination of the $|v_j\rangle$, i.e. $|b\rangle = \sum_{i=0}^{N-1} \beta_i |v_i\rangle$. Note that $\beta_i = \langle v_i | b \rangle$. Hence we get

$$|x\rangle = \sum_{i=0}^{N-1} \frac{1}{\lambda_i} \beta_i |v_i\rangle,$$

meaning that we can obtain the desired solution by computing the coordinates of b in the eigenbasis and inverting the eigenvalues.

In the classical framework this is too expensive since it requires the computation of the eigenvectors and eigenvalue which is in general very costly.

6.1.2 Description of the quantum algorithm

Transposing the idea above to the quantum paradigm, we need to use an unitary matrix to encode A . Since A is hermitian we can use the exponential operator and encode A as $U = e^{2\pi i A}$, which is indeed unitary. The following mathematical properties hold:

- If A is hermitian, matrix $U = e^{2\pi i A}$ is unitary.
- U has the same eigenvectors of A
- the eigenvalues of U are $e^{2\pi i \lambda_j}$, where λ_j are the eigenvalues of A .

We can then apply QPE to retrieve in the basis the binary representation of the λ_j , for the moment assume that λ_j can be exactly be represented with d digits. If we apply QPE with matrix U to vector $|b\rangle$ we have

$$\text{QPE}(|0\rangle^{\otimes d} |b\rangle) = \sum_{j=0}^{N-1} \beta_j \text{QPE}(|0\rangle^{\otimes d} |v_j\rangle) = \sum_{j=0}^{N-1} \beta_j |\lambda_j\rangle |v_j\rangle.$$

Note that $A^{-1}|b\rangle = \sum_{j=0}^{N-1} \frac{\beta_j}{\lambda_j} |v_j\rangle$ is proportional to $|x\rangle$. We need then to use $|\lambda_j\rangle$ to compute the factor λ_j^{-1} and multiply each of it by β_j . We need an additional ancilla qubit and a controlled rotation U_{cr} such that

$$U_{cr}|0\rangle|\lambda_j\rangle = \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right) |\lambda_j\rangle,$$

where $\tilde{\lambda}_j$ is an approximation of λ_j .

At this point we apply the uncomputation of the register where λ_j is maintained to bring it back to $|0\rangle^{\otimes d}$. This correspond to performing an inverse QPE. Indeed $\text{QPE}^\dagger(|\lambda_j\rangle|v_j\rangle) = |0\rangle^{\otimes d}|v_j\rangle$. Discarding the d ancilla we get that the final unitary U_{HHL} is such that

$$U_{HHL}(|0\rangle|b\rangle) = \sum_{j=0}^{N-1} \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right) \beta_j |v_j\rangle.$$

We then measure the ancilla qubit and postprocessing on the outcome 1, we get the vector

$$\tilde{x} = \sum_{j=0}^{N-1} \frac{C\beta_j}{\tilde{\lambda}_j} |v_j\rangle,$$

the normalize quantum state is $|\tilde{x}\rangle = \frac{\tilde{x}}{\|\tilde{x}\|}$.

The normalization constant can be obtained from the probability of measuring the ancilla at 1 and is $\text{prob}(1) = \|\sum_{j=0}^{N-1} \frac{C\beta_j}{\tilde{\lambda}_j} |v_j\rangle\|^2 = \|\tilde{x}\|^2$. To estimate such a probability we need to repeat the circuit many times.

6.1.3 Computational complexity

We have seen that C doesn't appear in the normalized final quantum state. However it affects the success probability. Indeed from

$$U_{HHL}|0\rangle|b\rangle = \sum_{j=0}^{N-1} \left(\sqrt{1 - \frac{C^2}{\tilde{\lambda}_j}} |0\rangle + \frac{C}{\tilde{\lambda}_j} |1\rangle \right) \beta_j |v_j\rangle,$$

we have that the probability of measuring the ancilla at 1 is proportional to the amplitude $\|CA^{-1}|b\rangle\|^2$. So to maximize the success probability, C should be chosen as large as possible, for example $C = \lambda_0$. In this case, since $\|A\| = \lambda_{N-1}$ and $\|A^{-1}\| = \lambda_0^{-1}$, the condition number is $\mu(A) = \lambda_{N-1}C^{-1}$. Since

$$\|A^{-1}|b\rangle\| \geq \frac{1}{\|A\|} \| |b\rangle \| = \lambda_{N-1}^{-1},$$

the probability of success is $\Omega(\mu(A)^{-2}\lambda_{N-1})$. Taking into account the complexity of QPE the query complexity of HHL is $O(\mu(A)^3/\varepsilon)$. We have a fast algorithm in the case U can be implemented with $\text{polylog}(N)$ gates, otherwise we do not have an exponential speedup. Moreover, note that there is a readout problem, because the normalized solution is in the state vector.

Bibliografia